

A journey through hyperbolic approximation of PDEs

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Journée de l'IMB - Edition 2025

Career preview

MCF - Bordeaux INP

Sep 2025 – Current

IMB Team CSM - INRIA Team Cardamom

Assistant Professor - University of Trento, Italy

Dec 2022 – Aug 2025

Team: Numerical analysis group of Michael Dumbser

Three years grant won through a competitive call of NextGenerationEU

Postdoctoral fellowship - University of Trento, Italy

Mar 2021 – Dec 2022

Team: Numerical analysis group of Michael Dumbser

Subject: A hyperbolic approximation of the Navier-Stokes-Korteweg equations.

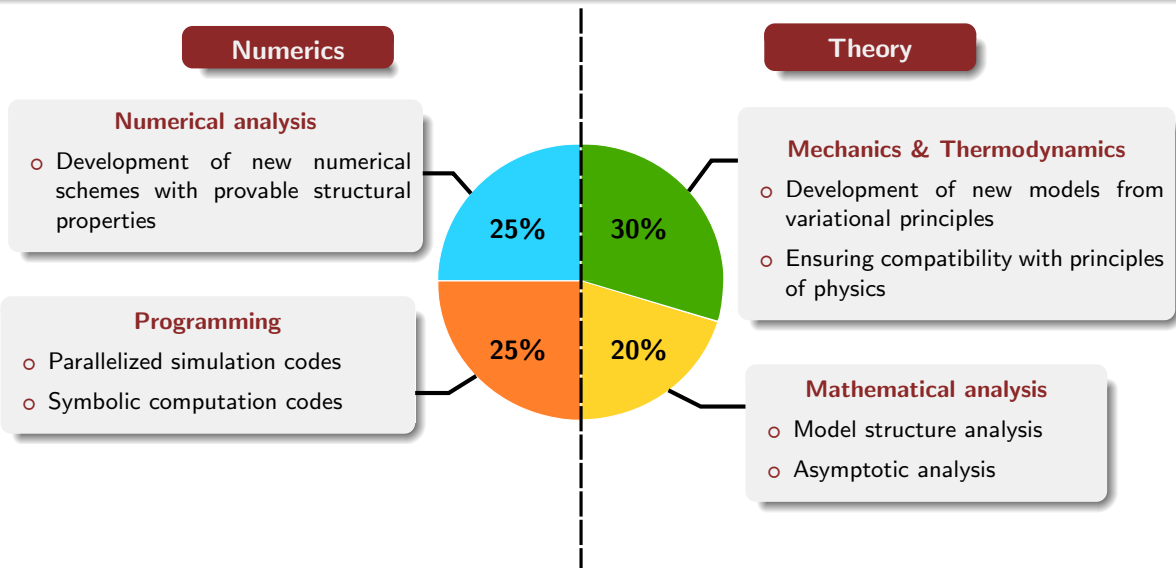
PhD Thesis - Université Paul Sabatier, Toulouse

Oct 2017 – Dec 2020

Advisors: Jean-Paul Vila (TLS), Nicolas Favrie and Sergey Gavrilyuk (AMU)

Subject: An augmented Lagrangian approach for Euler-Korteweg type equations.

Research profile



Main research motivation

1) Develop new models to describe **complex phenomena** by means of **hyperbolic¹ PDEs²**, that are compatible with the **principles of physics**.

1d case:
$$\partial_t \mathbf{Q} + \mathbf{A}(\mathbf{Q}) \cdot \partial_x \mathbf{Q} = \mathbf{S}(\mathbf{Q}), \quad \mathbf{Q} \in \mathbb{R}^n, \mathbf{A} : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times n} \quad (1)$$

2) Design new numerical methods, able to **solve these PDEs efficiently** while **preserving their mathematical structure** at the numerical level.

¹Eq (1) is hyperbolic if $\mathbf{A}(\mathbf{Q})$ is diagonalizable in $\mathbb{R}$, for any admissible $\mathbf{Q} \in \mathbb{R}^n$.

²PDE = Partial Differential Equations

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Why Hyperbolic PDEs ?

- ✓ The Cauchy problem for Symmetric-Hyperbolic PDEs is **locally well-posed**.
- ✓ Respects the principle of causality (**Everything travels at bounded speeds**).
- ✓ Benefits from a **rich literature of associated efficient numerical methods**.

¹Eq (1) is hyperbolic if $\mathbf{A}(\mathbf{Q})$ is diagonalizable in $\mathbb{R}$, for any admissible $\mathbf{Q} \in \mathbb{R}^n$.

²PDE = Partial Differential Equations

Existential question

What happens if a PDE we like is not hyperbolic ?

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What happens if a PDE we like is not hyperbolic ?
We just "make" it hyperbolic.

Cattaneo's model of relativistic heat transfer

Hyperbolic heat conduction equation [Cattaneo 1958].

$$u_t = \alpha u_{xx} \quad \Rightarrow \quad \begin{cases} u_t = q_x \\ q_t = (u_x - q)/\tau \end{cases} \quad \tau \ll 1$$

where $\alpha > 0$.

One expects to recover the original heat equation when $\tau \rightarrow 0$.

Hyperbolic approximations of dispersive equations

The Euler-Korteweg system

$$\begin{cases} \partial_t \rho + \operatorname{div}(\rho \mathbf{u}) = 0 & \text{(Mass cons.)} \\ \underbrace{\partial_t(\rho \mathbf{u}) + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u}) + \nabla p(\rho)}_{\text{Barotropic Euler part}} = \underbrace{\rho \nabla \left(K(\rho) \Delta \rho + \frac{1}{2} K'(\rho) |\nabla \rho|^2 \right)}_{\text{Korteweg stress}} & \text{(Momentum cons.)} \end{cases}$$

where $\rho = \rho(\mathbf{x}, t)$, $\mathbf{u} = \mathbf{u}(\mathbf{x}, t)$ and $(\mathbf{x}, t) \in \mathbb{R}^d \times [0, T]$

- ✓ Generalizes the Euler equations to dispersive fluid flows (waves, **surface tension**, etc ...)
- ✗ Right-hand side is difficult to handle numerically, slows down simulations.

drop.jpg

leaf.jpg

ripple.jpg

Lagrangian for the Euler-Korteweg system

(EK) system can be derived from the Lagrangian :

$$\mathcal{L} = \int_{\Omega_t} \left(\frac{\rho |\mathbf{u}|^2}{2} - W(\rho) - \gamma \frac{|\nabla \rho|^2}{2} \right) d\Omega$$

Variational principle
 +
 Differential constraint : $\rho_t + \operatorname{div}(\rho \mathbf{u}) = 0$

↓

$$(\rho \mathbf{u})_t + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u}) + \nabla(p(\rho)) = \gamma \rho \nabla(\Delta \rho)$$

with $p(\rho) = \rho W(\rho) - W(\rho)$

Augmented Lagrangian approach in a nutshell

Main idea

Transform a complicated PDE system into an approximate hyperbolic one:

✓ **Better mathematical properties**

✓ **Easier/faster to solve numerically.**

Original Lagrangian

$$\mathcal{L} = \int \left(\frac{\rho |\mathbf{u}|^2}{2} - f(\rho) - \gamma \frac{|\nabla \rho|^2}{2} \right) d\Omega$$

Penalty Method
+ New variables

Modified Lagrangian

$$\tilde{\mathcal{L}}_{\alpha,\beta} = \int \left(\rho \frac{|\mathbf{u}|^2}{2} - f(\rho) - \frac{\gamma}{2} |\nabla \eta|^2 - \frac{1}{2\alpha\rho} (\rho - \eta)^2 + \frac{\beta\rho}{2} \dot{\eta}^2 \right) d\Omega$$

Variational principle

First-order hyperbolic system

Change of variables

New Equations of Motion

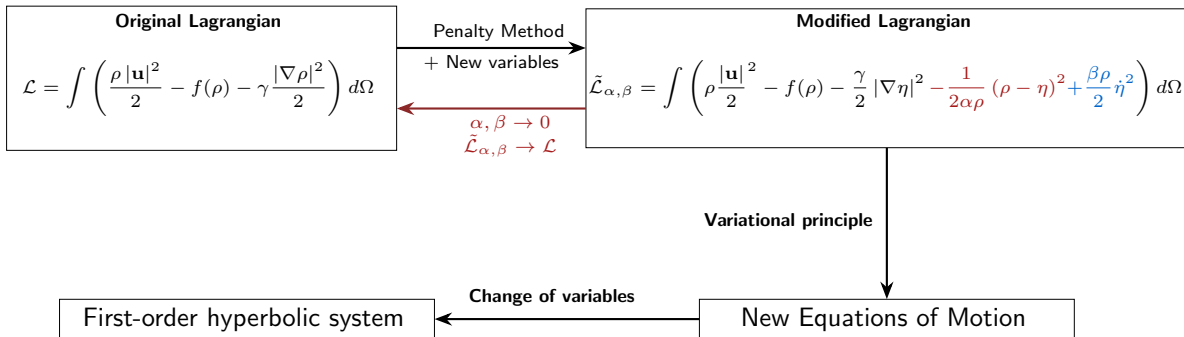
Augmented Lagrangian approach in a nutshell

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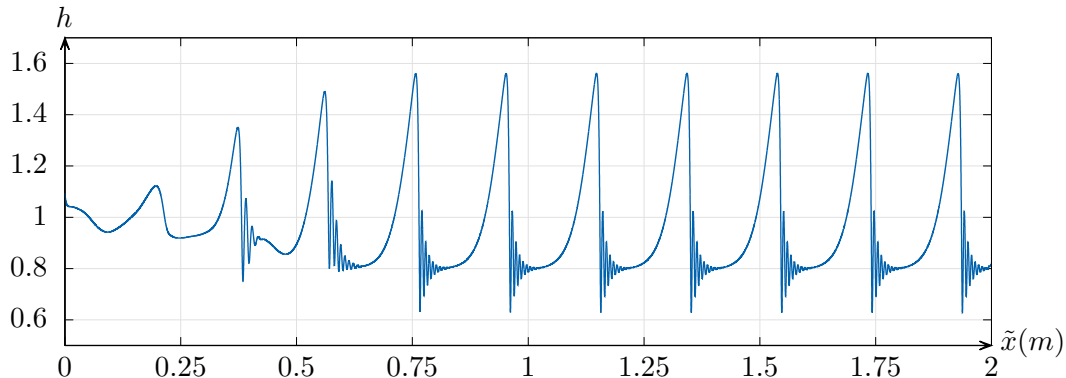
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Example - Thin film flows: Liu & Gollub's experiment (1994)



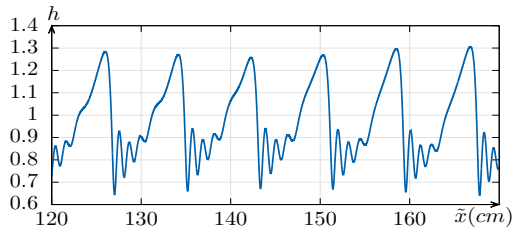
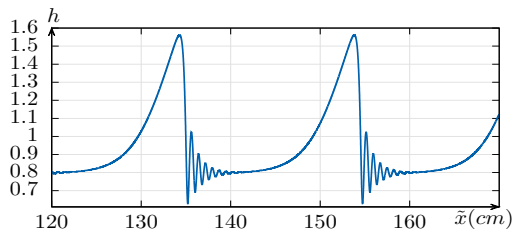
Dimensionless water height as a function of space.

👉 Numerical result obtained through a **Fortran/MPI code**.

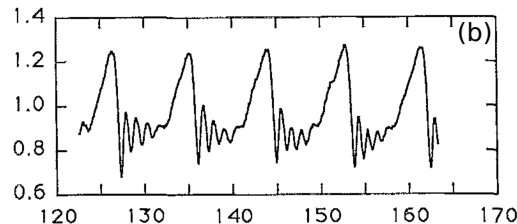
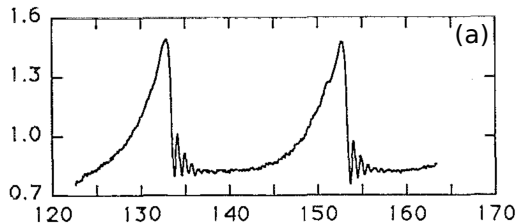
[6] **Dhaouadi F**, Gavriluk S, Vila JP. 2022 Hyperbolic relaxation models for thin films down an inclined plane. **Applied Mathematics and Computation** 433.

Example - Thin film flows: Liu & Gollub's experiment (1994)

Numerical simulation result



Experimental result



Inlet oscillation frequency : (a) 1.5Hz, (b) 3.0 Hz

Extension to the Navier-Stokes-Korteweg equations

$$\begin{aligned}\partial_t \rho + \operatorname{div}(\rho \mathbf{u}) &= 0 \\ \partial_t(\rho \mathbf{u}) + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u}) + \nabla p(\rho) &= \rho \nabla \left(K(\rho) \Delta \rho + \frac{1}{2} K'(\rho) |\nabla \rho|^2 \right) \\ &\quad + \mu \operatorname{div} \left(\nabla \mathbf{u} + \nabla \mathbf{u}^T - \frac{2}{3} \operatorname{div}(\mathbf{u}) \mathbf{I} \right)\end{aligned}$$

Hyperbolic NSK = **Hyperbolic EK** + **Hyperbolic NS**
 proposed by me done in Trento

images/Ostwald_2D_rho-2.pdf

👍 Results obtained by implementing my system in the Trento research team's code AMR3D (Fortran F90+MPI).

[5] **Dhaouadi F**, Dumbser M. 2022 A first order hyperbolic reformulation of the Navier-Stokes-Korteweg system based on the GPR model and an augmented Lagrangian approach. *Journal of Computational Physics* 470

Structure-Preserving numerical methods

Definition

Any numerical method designed to preserve a geometric, algebraic or physical constraint of the continuous PDE at the discrete level.

Examples:

- Positivity-preserving (Temperature or density must be non-negative,)
- domain-invariance (mass fraction must remain in $[0, 1]$, if initially so)
- **div/curl constraints** (divergence or curl of vectors must be zero or some value), etc.

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Involutive VS non-involutive constraints

Incompressible Euler

$$\begin{aligned}\nabla \cdot \mathbf{u} &= 0, \\ \mathbf{u}_t + (\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p &= 0.\end{aligned}$$

Maxwell equation for $\mathbf{B}$

$$\begin{aligned}\mathbf{B}_t + \nabla \times \mathbf{E} &= 0, \\ \nabla \cdot \mathbf{B} &= 0.\end{aligned}$$

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Maxwell equation for $\mathbf{B}$

$$\begin{aligned}\mathbf{B}_t + \nabla \times \mathbf{E} &= 0, \quad \Rightarrow (\nabla \cdot \mathbf{B})_t = 0 \\ \nabla \cdot \mathbf{B} &= 0.\end{aligned}$$

On the importance of structure-preserving schemes

Curl-free involution

The gradient field $\mathbf{p}$ obeys an equation of the form

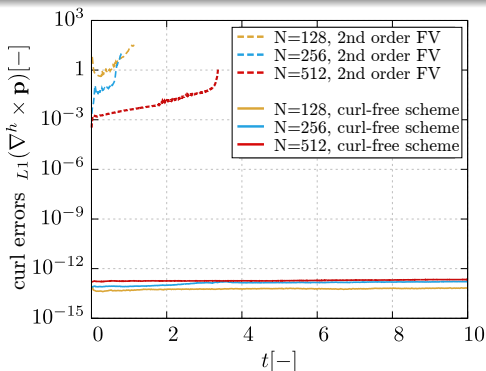
$$\partial_t \mathbf{p} = \nabla \phi$$

As a consequence:

if $\nabla \times \mathbf{p} = 0$ initially, it remains zero $\forall t \geq 0$

✓ Always true at the continuous level.

✗ Easily violated at the discrete level.



⇒ Non curl-free simulations **crash in finite time**.

👉 Results obtained by implementing our system and numerical method in the Trento research team's code OSIRIS.

[4] **Dhaouadi F**, Dumbser M. 2023 A structure-preserving finite volume scheme for a hyperbolic reformulation of the Navier–Stokes–Korteweg equations. *Mathematics* **11**, 876.

Extension 2: The Cahn-Hilliard equation (1958)

The Cahn-Hilliard equation writes

$$\frac{\partial c}{\partial t} = \Delta (c^3 - c - \gamma \Delta c) .$$

- $c \in [-1, 1]$ is the order parameter indicating the phases.
- $\gamma \ll 1$ is such that $\sqrt{\gamma}$ is the diffuse interface characteristic length.
- describes phase separation in binary systems: spinodal decomposition, Ostwald Ripening phenomena, etc.

Applications of Cahn-Hilliard equations

1) Simple model describing phase separation processes in binary mixtures.

2) Modeling of air pollution (Mixtures of gas/smoke)



3) Sedimentation problems

sedi.png

Hyperbolic Approximation of Cahn-Hilliard equation

After long and tedious procedures ... the PDE $c_t = \Delta (c^3 - c - \gamma \Delta c)$ becomes

$$\frac{\partial c}{\partial t} + \operatorname{div} \left(\frac{1}{\tau} \mathbf{q} \right) = 0$$

$$\frac{\partial \mathbf{q}}{\partial t} + \nabla (c^3 - c + \alpha(c - \varphi)) = -\frac{1}{\tau} \mathbf{q}$$

$$\frac{\partial w}{\partial t} - \operatorname{div} (\gamma \mathbf{p}) = -\alpha(\varphi - c)$$

$$\frac{\partial \mathbf{p}}{\partial t} - \frac{1}{\beta} \nabla w = 0$$

$$\frac{\partial \varphi}{\partial t} = \frac{1}{\beta} w$$

- System of hyperbolic equations with relaxations.
- Equations are conservative also in multiple dimensions.
- Also has a Lyapunov functional similarly to the original equation

Spinodal decomposition ($\gamma = 10^{-3}, \beta = 10^{-7}, \alpha = 500, \tau = 10^{-5}$)

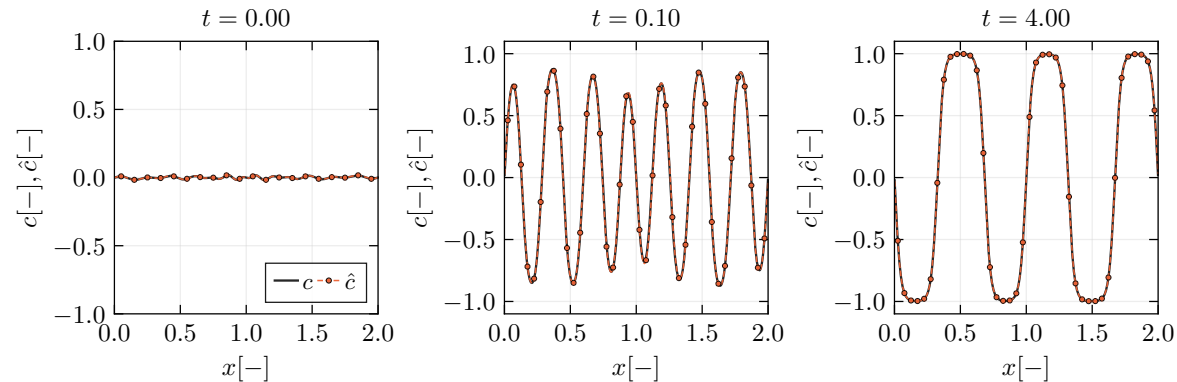


Figure 1: Comparison of the numerical results between the original model (orange) and its hyperbolic counterpart (black). $N = 2000$ computational cells.

Spinodal decomposition - Lyapunov function evolution ($\simeq 6b$ iterations)

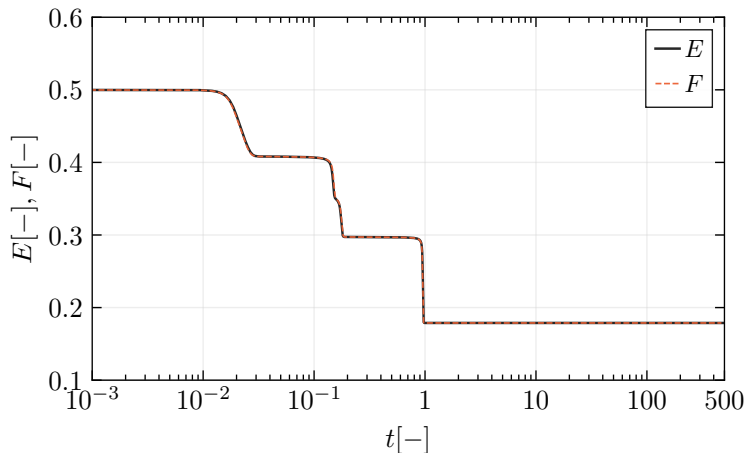


Figure 2: Lyapunov function evolution over time (F :original, E : hyperbolic.)

Spinodal decomposition - Lyapunov function evolution ($\simeq 30b$ iterations)

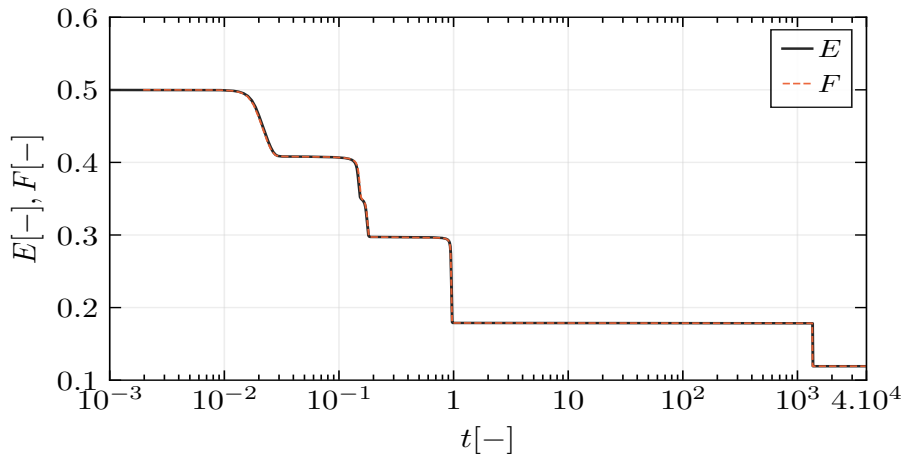


Figure 3: Lyapunov function evolution over time (F :original, E : hyperbolic.)

Ostwald Ripening in 1D ($\gamma = 10^{-3}, \beta = 10^{-7}, \alpha = 500, \tau = 10^{-4}$)

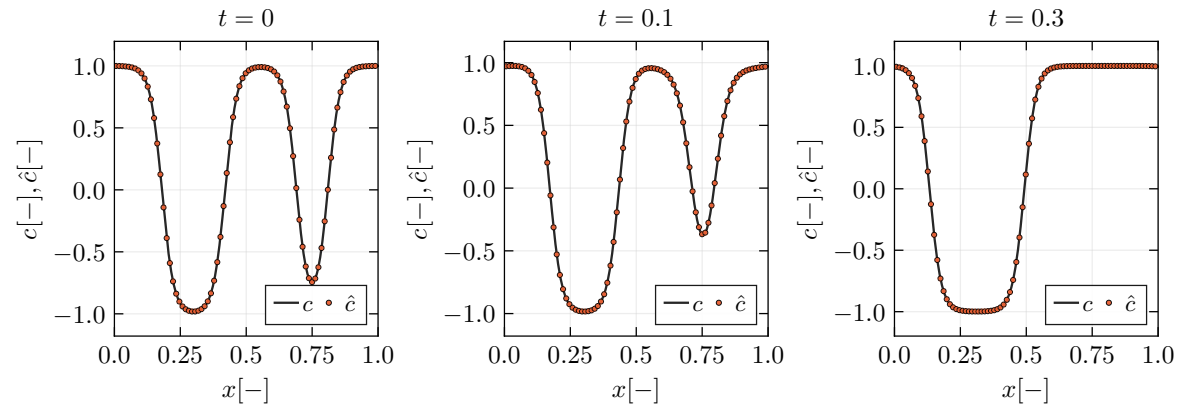


Figure 4: Comparison of the numerical solutions for hyperbolic Cahn-Hilliard model (black line) and the original model (red dots) for the Ostwald Ripening test case at times $t = \{0, 0.1, 0.3\}$.

Ostwald Ripening in 2D($\gamma = 10^{-3}, \beta = 10^{-7}, \alpha = 500, \tau = 10^{-5}$)

Ostwald Ripening in 2D : horizontal Cuts

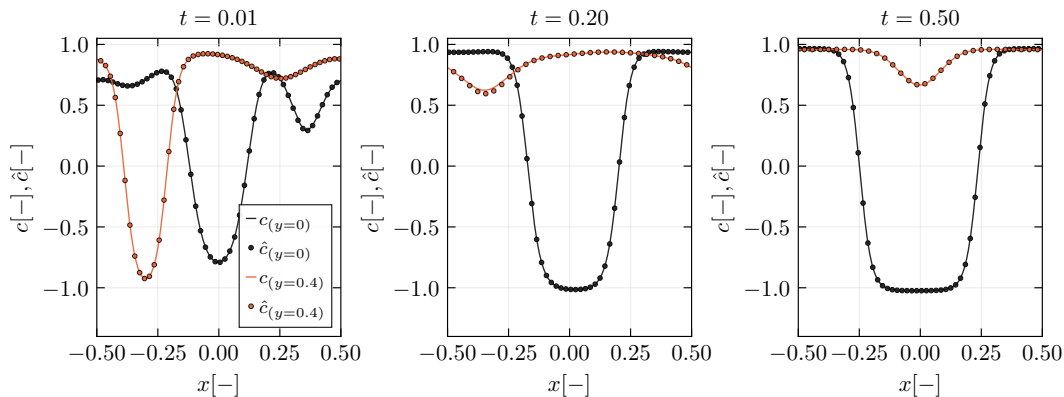


Figure 5: Horizontal cuts over the lines $y = 0$ (red) and $y = 0.4$ (black). Domain is 600×720

Two-way transfer of knowledge

What I can offer:

- Hyperbolic approximation of nonlinear system of PDEs.
- Systems of PDEs that are also consistent with the principles of thermodynamics.
- Analysis of hyperbolic models (Riemann invariants, nature of fields, symmetrization, etc)
- Knowledge on structure-preserving numerical methods.
- Symbolic code development using Wolfram Mathematica (Pipelines from equation to code/tex)

Subset of open questions / Aspects to analyze deeper:

Analysis of convergence of the solutions in the relaxation limit.

Extension from local to global well-posedness of solutions ?

Further analysis of compatible initial/boundary conditions

Thank you for your attention!

- [1] Dhaouadi F, Dumbser M, Gavrilyuk S. 2025 A first-order hyperbolic reformulation of the Cahn-Hilliard equation. **Proceedings of the Royal Society A**, 20240606. (10.1098/rspa.2024.0606)
- [2] Río-Martín L, Dhaouadi F, Dumbser M. 2025 An Exactly Curl-Free Finite-Volume/Finite-Difference Scheme for a Hyperbolic Compressible Isentropic Two-Phase Model. **Journal of Scientific Computing** 102, 13. (10.1007/s10915-024-02733-9)
- [3] Dhaouadi F, Gavrilyuk S. 2024 An Eulerian hyperbolic model for heat transfer derived via Hamilton's principle: analytical and numerical study. **Proceedings of the Royal Society A** 480, 20230440 . (10.1098/rspa.2023.0440)
- [4] Dhaouadi F, Dumbser M. 2023 A structure-preserving finite volume scheme for a hyperbolic reformulation of the Navier–Stokes–Korteweg equations. *Mathematics* 11, 876. (10.3390/math11040876)
- [5] Dhaouadi F, Dumbser M. 2022 A first order hyperbolic reformulation of the Navier-Stokes-Korteweg system based on the GPR model and an augmented Lagrangian approach. **Journal of Computational Physics** 470, 111544. (10.1016/j.jcp.2022.111544)
- [6] Dhaouadi F, Gavrilyuk S, Vila JP. 2022 Hyperbolic relaxation models for thin films down an inclined plane. **Applied Mathematics and Computation** 433, 127378. (10.1016/j.amc.2022.127378)
- [7] Dhaouadi F, Favrie N, Gavrilyuk S. 2019 Extended Lagrangian approach for the defocusing nonlinear Schrödinger equation. **Studies in Applied Mathematics** 142, 336–358. (10.1111/sapm.12238)

Appendix

Hints on calculus of variations (For general $K(\rho)$)

$$\tilde{\mathcal{L}} = \int_{\Omega_t} \left(\rho \frac{|\mathbf{u}|^2}{2} - W(\rho) - K(\rho) \frac{|\nabla \eta|^2}{2} - \frac{\rho}{2\alpha} \left(\frac{\eta}{\rho} - 1 \right)^2 \right) d\Omega$$

$$\tilde{\mathcal{L}}(\underbrace{\mathbf{u}, \rho}_{\delta \mathbf{x}}, \underbrace{\eta, \nabla \eta}_{\delta \eta}) \Rightarrow \text{Two Euler-Lagrange equations}$$

Virtual displacement of the continuum ($\delta \mathbf{x}$):

$$\begin{aligned} & (\rho \mathbf{u})_t + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u}) + \nabla(\rho W'(\rho) - W(\rho)) \\ &= -\operatorname{div}(K(\rho) \nabla \eta \otimes \nabla \eta) - \nabla \left(\frac{1}{2}(\rho K'(\rho) - K(\rho)) |\nabla \eta|^2 + \frac{\eta}{\alpha} \left(1 - \frac{\eta}{\rho} \right) \right) \end{aligned}$$

η variation ($\delta \eta$):

$$\frac{1}{\alpha} \left(1 - \frac{\eta}{\rho} \right) = - (K(\rho) \Delta \eta + K'(\rho) \nabla \rho \cdot \nabla \eta)$$

Eigenvalues - Hyperbolicity of the reformulated NSK system

$\Rightarrow$ 21 Eigenvalues (Linearized around $A = \mathbf{I}, \mathbf{p} = (p_1, 0, 0)^T$)

Transport: $\lambda_{1-9} = u_1,$

shear waves: $\begin{cases} \lambda_{10-11} = u_1 + c_s, \\ \lambda_{12-13} = u_1 - c_s, \end{cases}$

Cleaning waves: $\begin{cases} \lambda_{14-15} = u_1 - \sqrt{\gamma/\rho} a_c, \\ \lambda_{16-17} = u_1 + \sqrt{\gamma/\rho} a_c, \end{cases}$

Mixed waves:

$$\begin{cases} \lambda_{18} = u_1 - \sqrt{Z_1 + Z_2} \\ \lambda_{19} = u_1 - \sqrt{Z_1 - Z_2} \\ \lambda_{20} = u_1 + \sqrt{Z_1 + Z_2} \\ \lambda_{21} = u_1 + \sqrt{Z_1 - Z_2} \end{cases}, \begin{cases} Z_1 = \frac{1}{2}(a_0^2 + a_s^2 + a_\gamma^2 + a_\alpha^2 + a_\beta^2), \\ Z_2 = \sqrt{Z_1^2 - a_\beta^2(a_0^2 + a_\alpha^2 + a_s^2)}, \\ a_0 = \sqrt{\rho W''(\rho)}, \quad a_s = \sqrt{\frac{4}{3}c_s^2} \\ a_\alpha = \frac{\eta}{\rho\sqrt{\alpha}}, \quad a_\beta = \sqrt{\frac{\gamma}{\beta\rho}}, \quad a_\gamma = \sqrt{\frac{\gamma}{\rho}p_1^2} \end{cases}$$

Scaling of relaxations

Representative characteristic velocities

$$\left\{ \begin{array}{l} \lambda_{18} = u_1 - \sqrt{Z_1 + Z_2} \\ \lambda_{19} = u_1 - \sqrt{Z_1 - Z_2} \\ \lambda_{20} = u_1 + \sqrt{Z_1 + Z_2} \\ \lambda_{21} = u_1 + \sqrt{Z_1 - Z_2} \end{array} \right\}, \left\{ \begin{array}{l} Z_1 = \frac{1}{2}(a_0^2 + a_s^2 + a_\gamma^2 + a_\alpha^2 + a_\beta^2), \\ Z_2 = \sqrt{Z_1^2 - a_\beta^2(a_0^2 + a_\alpha^2 + a_s^2)}, \\ a_0 = \sqrt{\rho W''(\rho)}, \quad a_s = \sqrt{\frac{4}{3}c_s^2} \\ a_\alpha = \frac{\eta}{\rho\sqrt{\alpha}}, \quad a_\beta = \sqrt{\frac{\gamma}{\beta\rho}}, \quad a_\gamma = \sqrt{\frac{\gamma}{\rho}p_1^2} \end{array} \right.$$

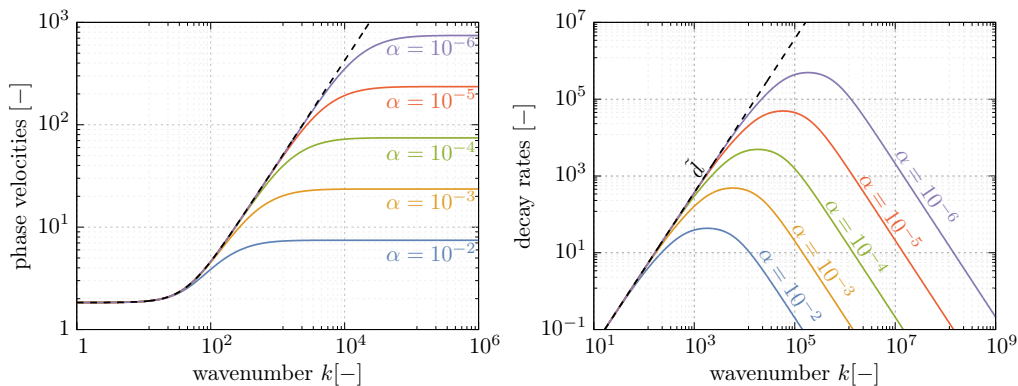
The different relaxation contributions scale as

$$a_\alpha^2 \sim \frac{1}{\alpha}, \quad a_\beta^2 \sim \frac{\gamma}{\beta\rho}, \quad a_s^2 \sim c_s^2$$

To keep the contributions at the same order of magnitude, we can take for example

$$\beta = \gamma\alpha, \quad c_s = \frac{1}{\sqrt{\alpha}}$$

Dispersion relation



Plot of the phase velocity (left) and the decay rate for several values of α along their counterparts for the Navier-Stokes-Korteweg system. The model parameters are as follows $\gamma = 10^{-3}$, $\mu = 10^{-3}$ and $\rho = 1.8$

System splitting

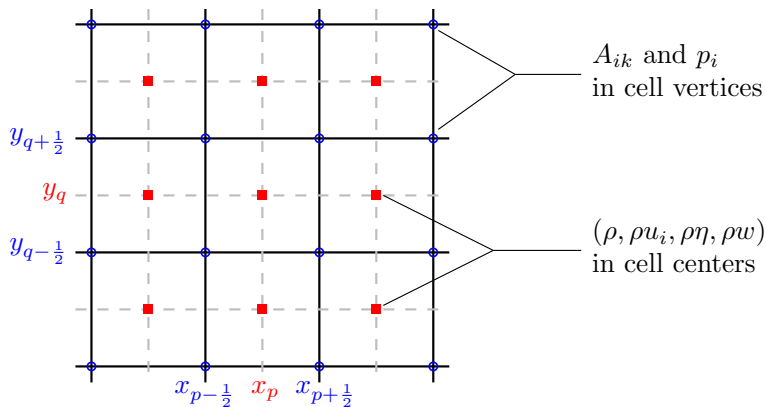
A set of classical conservation laws: **MUSCL-Hancock FV scheme**

$$\begin{aligned}\partial_t(\rho) &+ \operatorname{div}(\rho \mathbf{u}) = 0 \\ \partial_t(\rho \mathbf{u}) &+ \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u} + P(\rho) \mathbf{Id} - K_\alpha - \sigma) = 0 \\ \partial_t(\rho \eta) &+ \operatorname{div}(\rho \eta \mathbf{u}) = \rho w \\ \partial_t(\rho w) &+ \operatorname{div}(\rho w \mathbf{u} - \gamma \mathbf{p} / \beta) = (\alpha \beta)^{-1} (1 - \eta / \rho)\end{aligned}$$

A set of potentially curl constrained vectors: **Curl-free treatment**

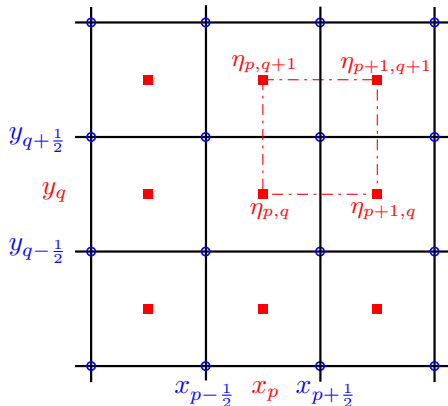
$$\begin{aligned}\partial_t(\mathbf{p}) &+ \nabla(\mathbf{p} \cdot \mathbf{u} - w) = 0, \\ \partial_t(\mathbf{A}_1) &+ \nabla(\mathbf{A}_1 \cdot \mathbf{u}) + (\nabla \times \mathbf{A}_1) \times \mathbf{u} = -\frac{1}{\tau} \mathbf{S}_1 \\ \partial_t(\mathbf{A}_2) &+ \nabla(\mathbf{A}_2 \cdot \mathbf{u}) + (\nabla \times \mathbf{A}_2) \times \mathbf{u} = -\frac{1}{\tau} \mathbf{S}_2 \\ \partial_t(\mathbf{A}_3) &+ \nabla(\mathbf{A}_3 \cdot \mathbf{u}) + (\nabla \times \mathbf{A}_3) \times \mathbf{u} = -\frac{1}{\tau} \mathbf{S}_3\end{aligned}$$

Exactly curl-free scheme: Staggered Grid



Schematic of the computational domain featuring the grid points and the staggered dual grid points. Red squares are barycenters and blue circles are the vertexes of the computational cells.

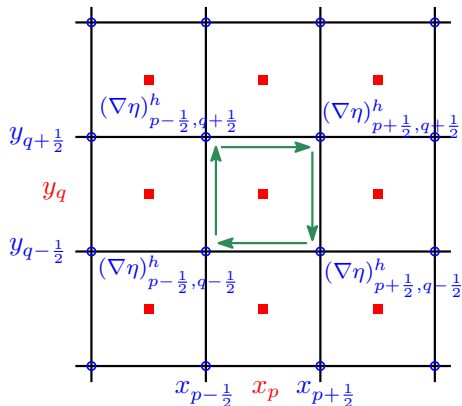
Exactly curl-free scheme: Gradient Stencil



$$\left\{ \begin{aligned} (\partial_x^h \phi)^{p+\frac{1}{2}, q+\frac{1}{2}} &= \frac{1}{2} \frac{\phi^{p+1, q} - \phi^{p, q}}{\Delta x} \\ &\quad + \frac{1}{2} \frac{\phi^{p+1, q+1} - \phi^{p, q+1}}{\Delta x}, \\ (\partial_y^h \phi)^{p+\frac{1}{2}, q+\frac{1}{2}} &= \frac{1}{2} \frac{\phi^{p, q+1} - \phi^{p, q}}{\Delta y} \\ &\quad + \frac{1}{2} \frac{\phi^{p+1, q+1} - \phi^{p+1, q}}{\Delta y}. \end{aligned} \right.$$

Stencil of the gradient field computed in every corner

Exactly curl-free scheme: Curl stencil



Stencil of the curl operator computed in every cell-center

Compatible discrete curl-operator

Based on this corner gradient, one can now define a compatible discrete curl operator such that $(\nabla^h \times \nabla^h \phi)^{p,q} \cdot \mathbf{e}_z$ is given by

$$\frac{(\partial_y^h \phi)^{p+\frac{1}{2},q+\frac{1}{2}} - (\partial_y^h \phi)^{p+\frac{1}{2},q-\frac{1}{2}}}{2\Delta x} + \frac{(\partial_y^h \phi)^{p-\frac{1}{2},q+\frac{1}{2}} - (\partial_y^h \phi)^{p-\frac{1}{2},q-\frac{1}{2}}}{2\Delta x} \\ - \frac{(\partial_x^h \phi)^{p+\frac{1}{2},q+\frac{1}{2}} - (\partial_x^h \phi)^{p-\frac{1}{2},q+\frac{1}{2}}}{2\Delta y} - \frac{(\partial_x^h \phi)^{p+\frac{1}{2},q-\frac{1}{2}} - (\partial_x^h \phi)^{p-\frac{1}{2},q-\frac{1}{2}}}{2\Delta y}.$$

It is straightforward to prove that

$$\nabla^h \times \nabla^h \phi \equiv 0$$

Update formulas ($h = \min(\Delta x, \Delta y)$)

For the conserved variables $\rho, \mathbf{u}, \rho\eta, \rho w$:

⇒ Classical MUSCL-Hancock scheme.

For the curl-free vector $\mathbf{p}$

$$p_k^{p+\frac{1}{2}, q+\frac{1}{2}, n+1} = p_k^{p+\frac{1}{2}, q+\frac{1}{2}, n} - \Delta t \nabla_k^h (p_j u_j - w - h \, c^* \nabla_j^h p_j)^{p+\frac{1}{2}, q+\frac{1}{2}, n}$$

Lastly, for $\mathbf{A}$

$$\begin{aligned} A_{ik}^{p+\frac{1}{2}, q+\frac{1}{2}, n+1} &= A_{ik}^{p+\frac{1}{2}, q+\frac{1}{2}, n} - \Delta t (\nabla_k^h (A_{ij} u_j) - h \, c^* \nabla_j^h A_{ij})^{p+\frac{1}{2}, q+\frac{1}{2}} \\ &\quad - \Delta t \, h \, c^* \varepsilon_{k j 3} \nabla_j^{p+\frac{1}{2}, q+\frac{1}{2}, n} (\varepsilon_{3 l m} \nabla_l^h A_{im}) \\ &\quad - \frac{\Delta t}{4} \sum_{r=0}^1 \sum_{s=0}^1 u_m^{p+r, q+s, n} \left((\nabla_m^h A_{ik})^{p+\frac{1}{2}, q+\frac{1}{2}} - (\nabla_k^h A_{im})^{p+\frac{1}{2}, q+\frac{1}{2}} \right) \\ &\quad - \Delta t \frac{1}{3\tau} \det(\mathbf{A}^{p+\frac{1}{2}, q+\frac{1}{2}, n+1})^{5/3} A_{im}^{p+\frac{1}{2}, q+\frac{1}{2}, n+1} \overset{\circ}{G}_{mk}^{p+\frac{1}{2}, q+\frac{1}{2}, n+1}. \end{aligned}$$

Wolfram Mathematica software ANALYTICA: Input example

$$\text{dims} = \begin{pmatrix} 1 \\ d \\ 1 \\ d \\ 1 \end{pmatrix} /. \text{unbracket}; (*\text{dimensions of the corresponding variables in V or Q} *)$$

$$V1 = \begin{pmatrix} c \\ q_i \\ w \\ p_i \\ \phi \end{pmatrix}; Q1 = \begin{pmatrix} c \\ q_i \\ w \\ p_i \\ \phi \end{pmatrix}; F1 = \begin{pmatrix} \frac{1}{\tau} q_k \\ (g'[c] + \alpha (c - \phi)) \delta_{i,k} \\ -\gamma p_k \\ \frac{-1}{\beta} w \delta_{i,k} \\ 0 \end{pmatrix};$$

$$B1 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}; S1 = \begin{pmatrix} 0 \\ -q_i / \tau \\ -\lambda (\phi - c) \\ 0 \\ w / \beta \end{pmatrix}$$

Wolfram Mathematica software ANALYTICA: Analysis output

1. System of equations

$$(1) \frac{\partial(c)}{\partial t} + \frac{\partial\left(\frac{q_1}{\tau}\right)}{\partial x} = 0$$

$$(2) \frac{\partial(q_1)}{\partial t} + \frac{\partial(\alpha(c - \phi) + g'(c))}{\partial x} = -\frac{q_1}{\tau}$$

$$(3) \frac{\partial(w)}{\partial t} + \frac{\partial(-\gamma p_1)}{\partial x} = -\lambda(\phi - c)$$

$$(4) \frac{\partial(p_1)}{\partial t} + \frac{\partial\left(-\frac{w}{\beta}\right)}{\partial x} = 0$$

$$(5) \frac{\partial(\phi)}{\partial t} = \frac{w}{\beta}$$

2. Jacobian Matrix

$$\mathbf{A} = \begin{pmatrix} & c & q_1 & w & p_1 & \phi \\ c & 0 & \frac{1}{\tau} & 0 & 0 & 0 \\ q_1 & \alpha + g''(c) & 0 & 0 & 0 & -\alpha \\ w & 0 & 0 & 0 & -\gamma & 0 \\ p_1 & 0 & 0 & -\frac{1}{\beta} & 0 & 0 \\ \phi & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad \{$$

2. Eigensystem Analysis

$$\xi_1 = 0$$

$$\left(\begin{array}{cccccc} \frac{\alpha}{\alpha + g''(c)} & 0 & 0 & 0 & 0 & 1 \end{array} \right)$$

=====

$$\xi_2 = -\frac{\sqrt{\gamma}}{\sqrt{\beta}}$$

$$\left(\begin{array}{cccccc} 0 & 0 & \sqrt{\beta} & \sqrt{\gamma} & 1 & 0 \end{array} \right)$$

=====

$$\xi_3 = \frac{\sqrt{\gamma}}{\sqrt{\beta}}$$

$$\left(\begin{array}{cccccc} 0 & 0 & -\sqrt{\beta} & \sqrt{\gamma} & 1 & 0 \end{array} \right)$$

=====

$$\xi_4 = -\frac{\sqrt{\alpha + g''[c]}}{\sqrt{\tau}}$$

$$\left(\begin{array}{cccccc} -\frac{1}{\sqrt{\tau} \sqrt{\alpha + g''(c)}} & 1 & 0 & 0 & 0 & 0 \end{array} \right)$$

=====

$$\xi_5 = \frac{\sqrt{\alpha + g''[c]}}{\sqrt{\tau}}$$

$$\left(\begin{array}{cccccc} \frac{1}{\sqrt{\tau} \sqrt{\alpha + g''(c)}} & 1 & 0 & 0 & 0 & 0 \end{array} \right)$$

Wolfram Mathematica software ANALYTICA: Code output

```
!=====
!===== STEP 0: AMRInit.f90 =====
!=====

#ifdef EQNTYPE5      !Allen-Cahn equation
| | nAux = ... ! Optional
#endif

!=====
!===== STEP 1: MainVariables.f90 =====
!=====

#ifdef EQNTYPE5      !Allen-Cahn equation
| INTEGER, PARAMETER :: nVar = 4
#endif

!===== STEP 2: Rusanov.f90 =====
Just add EQNTYPE5 to the list of EQNTYPES already there

!=====
!===== STEP 3: PDE.f90 fluxes =====
!===== EXERT CAUTION HERE!! CHECK EVERY LINE =====

#ifdef EQNTYPE5      !Allen-Cahn equation
|
| CALL PDECons2Prim(V,Q,x,time,iErr)
| F(1) = V(1) + Sqrt(alpha)*V(3)    ! u
| F(2) = -((gamma*\[Epsilon]**2*V(4))/beta)    ! w
| F(3) = Sqrt(alpha)*V(1) + alpha*V(3)    ! psi
| F(4) = -V(2)    ! p
|
#endif

!=====
```

Wolfram Mathematica software ANALYTICA: Code output

```
!=====
!===== PDE.f90 Source Terms =====
!=====

#ifdef EQNTYPE5      !Allen-Cahn equation
|
| S(2) = (Sqrt(EQN%alpha)*v(3))/EQN%beta  !w
| S(3) = -(Sqrt(EQN%alpha)*v(2))  !psi
|
#endif

!=====
!===== PDE.f90 Eigenvalues =====
!=====

#ifdef EQNTYPE5      !Allen-Cahn equation
|
| L(1) = 0.0
| L(2) = 1 + EQN%alpha
| L(3) = -((Sqrt(EQN%gamma)*EQN%\[Epsilon])/Sqrt(EQN%beta))
| L(4) = (Sqrt(EQN%gamma)*EQN%\[Epsilon])/Sqrt(EQN%beta)
|
#endif

!=====
!===== PDE.f90 Prim2Cons =====
!=====

#ifdef EQNTYPE5      !Allen-Cahn equation
|
| Q(1) = V(1)  ! u
| Q(2) = V(2)  ! w
| Q(3) = V(3)  ! psi
```